

# Vitrek V71 HiPot Controller

A Python/Flask web application for controlling the **Vitrek V71 Hi-Pot Tester** over USB (HID-to-UART) or RS-232. Runs test sequences, stores results in a local SQLite database, and exports to Excel for SharePoint sync. Built at [Juniper Design](#).

 **Print-ready PDF:** `docs/pdf/README.pdf`

## Features

- **USB and RS-232 support** — USB via Silicon Labs CP2110 HID-to-UART DLL (SLABHIDtoUART.dll); RS-232 via pyserial
- **Full V7X command set** — ACW, DCW, IR, GB, CONT test steps; sequence management; live measurements
- **Web UI** — browser-based interface, no installation needed on operator machines (just open `http://localhost:5000`)
- **SQLite results database** — every test session and step result persisted locally
- **Excel export** — formatted, color-coded workbook per session or bulk; ready for SharePoint

## Quick Start

### 1. Install dependencies

```
pip install -r requirements.txt
```

### 2. Connect the V71

- **USB:** Set `CONFIG MENU → INTERFACE = USB` on the V71 front panel. Connect the USB-B cable.
- **RS-232:** Set `CONFIG MENU → INTERFACE = RS232` and matching baud rate. Use a 9-wire null-modem cable.

### 3. Run the app

```
python app.py
```

Open `http://localhost:5000` in your browser.

# Project Structure

- └─ app.py                    Flask web app + REST API + embedded UI
- └─ v71\_driver.py           Low-level USB/serial driver (ctypes + pyserial)
- └─ database.py             SQLite schema and CRUD
- └─ excel\_export.py         openpyxl Excel workbook generator
- └─ requirements.txt
- └─ hipot\_results.db        SQLite database (auto-created on first run)
- └─ docs/
  - └─ V7x\_Series\_Operating\_Manual.pdf

## REST API

| Method | Endpoint          | Description                                                                                  |
|--------|-------------------|----------------------------------------------------------------------------------------------|
| POST   | /api/connect      | Connect to V71 ( { "mode": "usb" } or { "mode": "serial", "port": "COM3", "baud": 115200 } ) |
| POST   | /api/disconnect   | Disconnect                                                                                   |
| GET    | /api/status       | Current connection + run state                                                               |
| POST   | /api/run          | Start a test sequence (see below)                                                            |
| POST   | /api/abort        | Abort running test                                                                           |
| POST   | /api/cont         | Continue from HOLD step                                                                      |
| GET    | /api/live         | Live VOLTS/AMPS/OHMS during a run                                                            |
| GET    | /api/sessions     | Recent test sessions                                                                         |
| GET    | /api/session/<id> | Session detail + step results                                                                |
| GET    | /api/export       | Download all sessions as Excel                                                               |
| GET    | /api/export/<id>  | Download single session as Excel                                                             |

## Example: run a test via API

```
POST /api/run
{
  "operator": "Jason",
  "part_number": "PCB-001",
  "serial_number": "SN-12345",
  "steps": [
    { "type": "ACW", "voltage": 1500, "ramp": 2, "dwell": 60, "max_leakage": 0.005 },
    { "type": "IR", "voltage": 500, "dwell": 60, "min_resistance": 100000000 },
    { "type": "GB", "current": 25, "dwell": 5, "max_ohm": 0.1 }
  ]
}
```

---

# USB Communication Details

---

The V71 uses a **Silicon Labs CP2110 HID-to-UART bridge**. Communication is handled through:

- SLABHIDtoUART.dll + SLABHIDDevice.dll (x64 versions from software/drivers/ )
- USB VID: 4292 (0x10C4), PID: 34869 (0x8835)
- UART config: 115200 baud, 8N1, RTS/CTS flow control
- Protocol: ASCII commands terminated with \r\n

**Key commands** (from Section 6 of the operating manual):

| Command         | Description                 |
|-----------------|-----------------------------|
| *IDN?           | Identify device             |
| *RST            | Reset / clear               |
| *ERR?           | Read error register         |
| NOSEQ           | Clear active sequence       |
| ADD,ACW,...     | Add ACW test step           |
| ADD,DCW,...     | Add DCW test step           |
| ADD,IR,...      | Add IR test step            |
| ADD,GB,...      | Add Ground Bond step        |
| ADD,CONT,...    | Add Continuity step         |
| RUN             | Start active sequence       |
| ABORT           | Abort running sequence      |
| RUN?            | Query if running (0/1)      |
| RSLT?           | Overall result bitmask      |
| STAT?           | Per-step pass/fail string   |
| STEPRSLT?,<n>   | Detailed results for step n |
| MEASRSLT?,AMPS  | Live current measurement    |
| MEASRSLT?,VOLTS | Live voltage measurement    |

---

## Excel Export

---

Each export workbook contains: - **Summary sheet** — all sessions with pass/fail, sortable/filterable - **Per-session sheets** — full step results with color coding (green=pass, red=fail)

The `.xlsx` file is suitable for direct upload or auto-sync to SharePoint.

---

## Notes

---

- The DLL must be accessible at `software/drivers/USB_DLLs_and_Headers/USB_DLLs_and_Headers/x64/`
- The app must be compiled/run as **x64** to match the provided DLL architecture
- After a DUT breakdown event on USB, the V71 may disconnect and reconnect — the driver detects this and surfaces an error; simply reconnect from the UI
- RS-232 requires hardware handshaking (RTS/CTS) — ensure your cable is fully wired (9-wire null modem)